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WEEK 5 - ReactJS Hands ON

Exercise 1

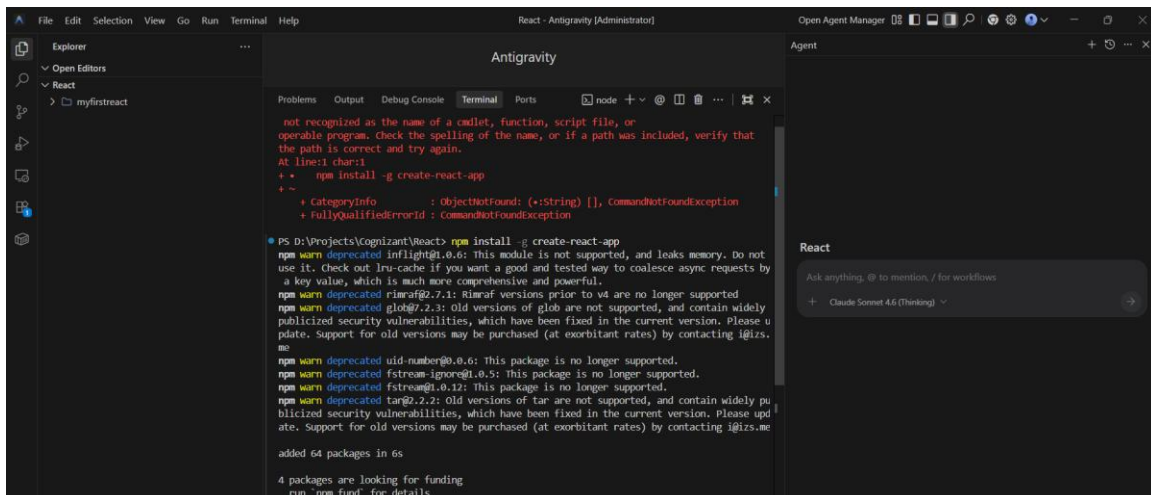
Create myfirstreact app and display 'Welcome to the first session of React'.

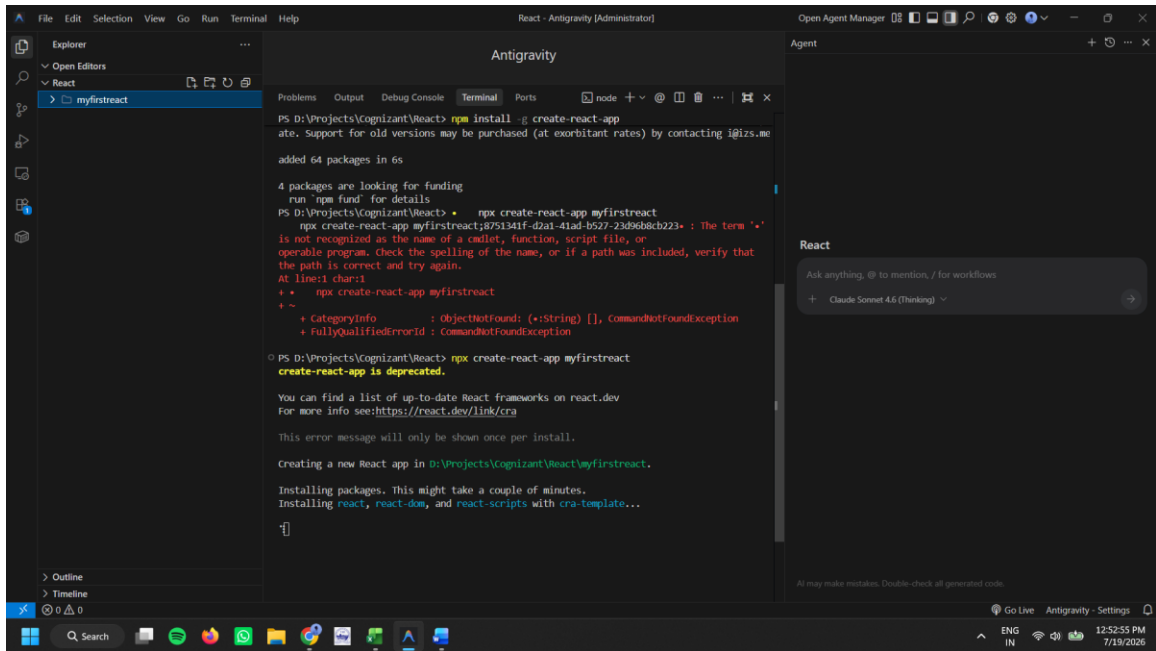
Commands

- npm install -g create-react-app
- npx create-react-app myfirstreact
- cd myfirstreact
- npm start

Solution:

```
function App() {  
  return <h1>Welcome to the first session of React</h1>;  
}  
export default App;
```





Exercise 2

Class Components

Aim:

StudentApp with Home, About and Contact components.

Commands

- `npx create-react-app StudentApp`
- `cd StudentApp`
- `npm start`

Sample Code

```
// Home.js
import React, {Component} from 'react';
export default class Home extends Component{
  render() {
    return <h3>Welcome to the Home Page of Student Management
Portal</h3>;
  }
}
```

Exercise 3

Functional Components

Aim:

Score Calculator using props and CSS.

Commands

- `npx create-react-app scorecalculatorapp`
- `npm start`

Sample Code

```
export function CalculateScore({Name, School, total, goal}) {
  const score = ((total/goal).toFixed(2));
  return (
    <div>
      <h1>Student Details</h1>
      <p>Name: {Name}</p>
      <p>School: {School}</p>
      <p>Total: {total}</p>
      <p>Score: {score}%</p>
    </div>
  )
}
```

```
);  
}
```

Exercise 4

Lifecycle Methods

Aim:

Create a React application that fetches blog posts using `componentDidMount()` and handles errors using `componentDidCatch()`.

Commands

- `npx create-react-app blogapp`
- `cd blogapp`
- `npm start`

Post.js

```
class Post {  
  
  constructor(id, title, body) {  
  
    this.id = id;  
  
    this.title = title;  
  
    this.body = body;  
  
  }  
  
}
```

```
export default Post;
```

Posts.js

```
import React, { Component } from "react";  
  
import Post from "../Post";
```

```
class Posts extends Component {  
  
  constructor(props) {
```

```

    super(props);

    this.state = { posts: [] };
  }

  loadPosts() {
    fetch("https://jsonplaceholder.typicode.com/posts")
      .then((res) => res.json())
      .then((data) => {
        const posts = data.map(
          (item) => new Post(item.id, item.title, item.body)
        );
        this.setState({ posts });
      });
  }

  componentDidMount() {
    this.loadPosts();
  }

  componentDidCatch(error) {
    alert("Something went wrong!");
    console.log(error);
  }

  render() {
    return (

```

```

    <div>
      <h1>Blog Posts</h1>
      {this.state.posts.map((post) => (
        <div key={post.id}>
          <h3>{post.title}</h3>
          <p>{post.body}</p>
          <hr />
        </div>
      ))}
    </div>
  );
}
}

```

```
export default Posts;
```

App.js

```
import Posts from "./Posts";
```

```

function App() {
  return <Posts />;
}

```

```
export default App;
```

Exercise 5

CSS Modules & Styling

Aim

Style the React application using **CSS Modules**. Create a CohortDetails.module.css file and apply styles to display ongoing and completed cohorts.

Commands

npm install

npm start

Folder Structure

src

|

└─ App.js

└─ CohortDetails.js

└─ CohortDetails.module.css

└─ index.js

CohortDetails.module.css

```
.box {  
  width: 300px;  
  display: inline-block;  
  margin: 10px;  
  padding: 10px 20px;  
  border: 1px solid black;  
  border-radius: 10px;
```

```
}
```

```
dt {
```

```
  font-weight: 500;
```

```
}
```

CohortDetails.js

```
import styles from "./CohortDetails.module.css";
```

```
function CohortDetails(props) {
```

```
  return (
```

```
    <div className={styles.box}>
```

```
      <h3
```

```
        style={{
```

```
          color:
```

```
            props.status === "ongoing"
```

```
              ? "green"
```

```
              : "blue"
```

```
        }}  
      >
```

```
        {props.name}
```

```
    </h3>
```



```
<dl>
```

```
  <dt>Started On</dt>
```

```
  <dd>{props.startDate}</dd>
```

```
  <dt>Current Status</dt>
```

```
  <dd>{props.status}</dd>
```

```
  <dt>Coach</dt>
```

```
  <dd>{props.coach}</dd>
```

```
  <dt>Trainer</dt>
```

```
  <dd>{props.trainer}</dd>
```

```
</dl>
```

```
</div>
```

```
);
```

```
}
```

```
export default CohortDetails;
```

App.js

```
import CohortDetails from "../CohortDetails";
```

```
function App() {  
  
  return (  
  
    <div>  
  
      <h1>My Academy Cohorts</h1>  
  
      <CohortDetails  
        name="React JS"  
        startDate="10-Jan-2025"  
        status="ongoing"  
        coach="John"  
        trainer="David"  
      />  
  
      <CohortDetails  
        name="Java Full Stack"  
        startDate="15-Dec-2024"  
        status="completed"  
        coach="Smith"  
        trainer="Alex"  
      />  
  
      <CohortDetails
```

```

        name="Python"
        startDate="20-Feb-2025"
        status="ongoing"
        coach="Emma"
        trainer="Sophia"
    />

</div>

);

}

export default App;

```

index.js

```

import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";

const root = ReactDOM.createRoot(
    document.getElementById("root")
);

root.render(
    <React.StrictMode>

```

```
    <App />  
  </React.StrictMode>  
);
```